Exercise 1

Submission Deadline:
10.12.2025, 11:55 pm

General Information

This exercise will guide you through various surface-EMG (sEMG) data processing and analysis steps. We suggest using MATLAB to visualize, quantify and process the EMG data and the corresponding force signals during the exercises. Note that using Python is also allowed, although we will use MATLAB for the exemplary solution. If you have not installed MATLAB yet, click on the link below and take advantage of FAU's Campus-Wide License to install and use MATLAB on your computer: www.mathworks.com/products/matlab/student.html

In addition to the base MATLAB version, you will likely need the MATLAB signal processing toolbox (depends on the functions you want to use). You can install it in MATLAB under Home -> Add-Ons.

MATLAB provides a lot of built-in functions that can make your life easier during these exercises. It is always a good idea to check the **MATLAB documentation**. Don´t hesitate to also use **Google**, **an AI language model** or **write in the StudOn forum**. Also check the "Requirements and Tips" section on the last page of this document.

Submission:

1. **Your code** as a MATLAB (.m) or Python (.py) file. For MATLAB it's mandatory to use the R2025b version. For python version 3.13. The code must be structured according to the tasks and generate all plots and figures required by the tasks. The plots and figures that must be generated by your code are **written in this colour**. Further requirements for the plots can be found in the "Requirements and Tips" section of this document. **You have to submit the code file inside of the StudOn online test (see next point).**
2. **The online test**. It can be accessed in the StudOn course from the time of the exercise presentation until the submission deadline. **The test can only be submitted once.** The test contains both single/multiple choice questions and written text questions. For questions that require written text as an answer, 1-3 sentences or bullet points should be enough. The first "question" of the test is reserved for the code submission, where you have to upload your .m or .py file.

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- **Only one code file may be uploaded**. The code must be written in such a way that it runs completely with a single click on "Run" and fulfills all tasks. If the code does not run due to any errors, the submission will be marked as failed.
- This is an **individual exercise**. If the submitted code is identical to another submission, both submissions will be marked as failed.
- Passing the exercise is only possible if both, code AND online test are submitted.

Note that the interpretation of the exercise results and the physiological concepts behind these **are relevant for the exam.** Additionally, there will be **bonus points for each successfully passed exercise.** The exercise has a pass mark of 70 % of maximum points (max: 27 pts).

The Dataset

Along with this document, you received one data file: ***wrist_flexion_extension.mat***. The file contains surface-EMG data acquired with an electrode bracelet containing 32 electrodes (Figure 1. B) worn on the skin of the proximal forearm. Two different recordings have been conducted.

In the first recording, the subject performed an isometric **wrist flexion**, trying to follow a predefined force level in the form of a ramp, with a plateau force amplitude of 40% of the subject's maximum force (maximum voluntary contraction (MVC)). See Figure 1. A for the experimental protocol.

In the second recording, the subject performed an isometric **wrist extension**. Again, the subject followed a force ramp, with the plateau phase at 40% of the subjects MVC in that direction.

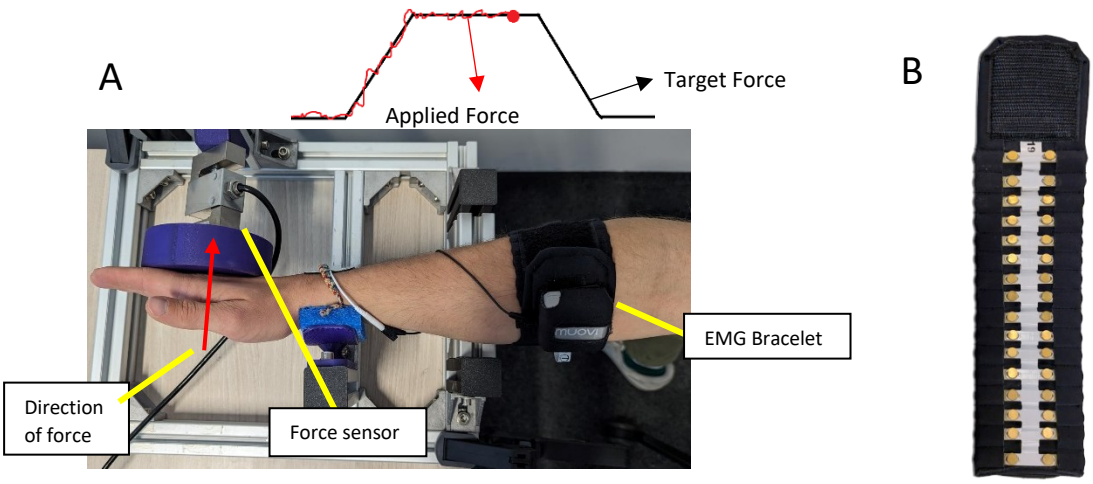

Figure 1. A: *Experimental protocol. Right hand is positioned in a wrist dynamometer with force transducer measuring wrist extension force. A 32-channel surface-EMG electrode bracelet records data from subject's proximal forearm muscles during isometric contractions. Target force ramp is visible on screen for the subject to follow.* ***B:*** *32 channel high-density surface-EMG electrode bracelet used in the data acquisition. The electrodes are aranged in a 16 x 2 matrix. Each electrode corresponds to one channel of EMG signal in the resulting dataset.*

The dataset contains the following relevant variables:

- **EMG signals (`EMG_raw_fl`, `EMG_raw_ex`):**
  The raw EMG signals in mV of all 32 channels stored inside a matrix with the shape *channels x samples*. EMG_raw_fl contains the flexion signal, EMG_raw_ex contains the extension signal.

- **Force signals (`force_fl`, `force_ex`)**
  The force signals measured by the force sensor in mV. force_fl contains the force signal of the flexion, force_ex contains the force signal of the extension.

- **Sampling frequency (`fsamp`)**
  Sampling frequency of EMG and force signals in Hz

Task 1. Data preprocessing & visualisation

Visualizing the signals is always a good idea to get a first impression and feeling for your data. Perform Tasks 1.1 – 1.6 for **both** recordings.

1.1 Convert the force transducer signal to Newtons using the device-specific conversion factor. $Force\ in\ N = \frac{Force}{ConversionFactor} * Gravity$ *ConversionFactor = 0.1; Gravity= 9.81 m/s²*

1.2 **Plot the force signal (2 pt)** in Newtons on the y-axis over time in seconds (invert the force signal to correct the negative values caused by the recording of the pressure direction). Include axis labels and units. Take a close look at the signal.

1.3 **Plot all EMG channels in mV over time listed one below the other in a single plot. Choose a suitable interspace between the plotted channels, so all of them are visible. (1 pt)**

1.4 Delete the channels from the EMG_raw signals that show strange behavior (e.g. no signal during some time, strong motion artifacts, ...) from both files. **Plot the remaining EMG channels again (as in task 1.3) to check if you deleted the correct channels. (1 pt)** **From now on, only work with the remaining channels**. *(Hint: Flexion has 2 channels to delete, extension only 1)*

1.5 The EMG signals contain noise that will cause problems in later analysis steps. Filter the EMG signal with a suitable filter. Take care to only remove the noise, but to preserve all valuable information in the data. **Plot the unfiltered AND filtered signal of one channel of your choice** (choose a channel with high activiation) **in the same plot, so the difference between the signals is visible (2 pts)** (choose a suitable visualisation window). In addition, **perform an FFT analysis of the unfiltered AND filtered EMG signal of only the first channel and plot them together in one figure (using two subplots) to see if the filter works (1 pt).**

1.6 **Plot an exemplary channel of your choice (choose a channel of high activation for each dataset) of the filtered EMG data together with the force signal in N in one figure (1 pt)** (use two seperate y-axes for that!).

Task 2. Force steadiness

Calculating the force steadiness during the plateau phase of the force generation is a basic first analysis of the force data. Perform Task 2.1 for **both** recordings.

2.1. **Compute the coefficient of variation (CV) (1 pt)** **during the plateau phase** of the force (you can select the start and end point of the plateau phase manually). The result will only be one value, so **don't plot it**, instead write it as an output (e.g.: "Task 2.1: Wrist Flexion, CV = 4.20 %") in the command window (Python: in the terminal).
The CV can be calculated by dividing the signals standard deviation by the signals mean, multiplied by 100 to get the value in %. $CV = \frac{Standard\ Deviation\ \sigma}{mean\ \bar{x}} * 100$

Task 3. Rate of force development

Here you compute the rate of force development (RFD). Perform Tasks 3.1 and 3.2 for **both** recordings.

3.1 Compute the RFD of the **whole force signal** using a moving window with a 1-sample step size. The window length should be 1000 ms (or 2000 samples, as the sampling frequency is 2000 Hz). Search how to calculate the RFD in the literature [1][2].

3.2 **Plot the RFD in N/s over time in seconds. (2 pts)**

Task 4. Root-Mean-Square (RMS) of EMG

The RMS is an important feature in EMG signal analysis that provides insight into the amplitude of the physiological activity by giving a measure of the signal power. It is calculated as the square root of the signal mean in a moving window. As you compute the square root, this also rectifies the signal. Perform Task 4.1 – 4.3 for **both** recordings.

4.1. Use the previously filtered EMG data (from Task 1.5) and compute the average across all remaining filtered channels, with an outcome shape of *1 x samples*.

4.2. Compute the RMS of this average as a moving average with a window length of 400 ms. **Visualize the result in a suitable way (2 pts).**

4.3. Visualize the correlation between the resulting RMS and the respective force signal by **plotting the normalized RMS (x-axis) against the normalized Force (y-axis) (2 pts)** in the form of a scatter plot. Also quantify the correlation by **computing the correlation coefficient r (1 pt)**. Put the correlation coefficient r in the title of your scatter plot (e.g.: Task 4.3: Wrist Flexion – Correlation between RMS and Force – r = 0.914)

Literature

Access the literature via the university VPN.

1. Comfort P, Allen M, Graham-Smith P. Comparisons of peak ground reaction force and rate of force development during variations of the power clean. J Strength Cond Res. 2011 May;25(5):1235-9. doi: 10.1519/JSC.0b013e3181d6dc0d.
2. Haff GG, Ruben RP, Lider J, Twine C, Cormie P. A comparison of methods for determining the rate of force development during isometric midthigh clean pulls. J Strength Cond Res. 2015 Feb;29(2):386-95. doi: 10.1519/JSC.0000000000000705.

Requirements and Tips

- **Code Structure:** Keep your script well-structured and clear. For this, please create script sections by writing `%%` (in MATLAB), followed by a blank space and a section title (please choose the section title according to the task). You can execute single script sections with the `Run Section` button in the toolbar or `Ctrl+Enter`, instead of always having to execute the entire script. (https://www.mathworks.com/help/matlab/matlab_prog/create-and-run-sections.html)

- **Plots and figures:** Each plot must have a clear title. The title should contain the respective task and a short description (e.g.: *Task 1.2: Force Signal of Isometric Ankle Dorsiflexion*). If a graph with axes is included, the axes must be labelled (with units and values).

  If you want to plot multiple related figures into one figure window, check out the `subplot` command. If you want to plot multiple plots into the same figure, use the `hold on` command. `yyaxis right` lets you plot multiple data series into one plot with independent y axes.

  (https://www.mathworks.com/help/matlab/ref/subplot.html)
  (https://www.mathworks.com/help/matlab/ref/hold.html)
  (https://www.mathworks.com/help/matlab/ref/yyaxis.html)

- **Custom Functions:** If you need to apply the same data processing steps to multiple sets of data, is it advised to write a custom function for these steps, that you can then pass the datasets to. In this exercise, please define your functions at the bottom of your code. Check https://www.mathworks.com/help/matlab/ref/function.html on how to define functions.

- If you do not know how to code a certain task, there is probably a **built-in MATLAB function** already available. Check the documentation, Google or the AI language model of your choice.